

AI ENGINEER TECHNICAL ASSIGNMENT – SYNTHETIC DATA GENERATION ENGINE (SDGE) Questions and Answers:

[See README.MD or Notebook.ipyn for full breakdown, and analysis. This is just an easy read format for questions posed]

Model rationale

WHY DID YOU CHOOSE DIFFUSION VS GAN?

Tabular cancer data mixes continuous genetic data with categorical clinical variables. Diffusion models handle mixed types naturally through score-based denoising compared to GAN. GANs may collapse and ignore minority classes, while diffusion models improve on training stability, coverage, and will synthesize better data.

WHY THIS SPECIFIC ARCHITECTURE VARIANT (E.G. DDPM VS DDIM, UNET DEPTH, NOISE SCHEDULER, ETC.)?

DDPM denoising better captures biological variability due to its fully stochastic nature. DDIM's deterministic sampling trades diversity for speed. For heterogeneous genomics, DDPM preserves richer distributional structure, synthesizing better data.

Known limitations & possible future improvements.

ALTHOUGH THE CURRENT DIFFUSION BASED SDGE PRODUCES REALISTIC AND DIVERSE DATA, THERE ARE SEVERAL LIMITATIONS:

- 1. Trade off with latent-only modeling: Model operates on PCA- reduced latent space rather than full expression, which may omit finer grain signals.
- 2. Better clinical data: Better patient data. A lot of clinical data was dropped or imputed.
- 3. Small sample size: TCGA-OV has usable sample size of 429 before preprocessing. This may limit the richness of the learned distribution. Some synthetic augmentation could help with this.

WHAT I WOULD DO WITH MORE TIME:

- 1. Architecture Enhancements:
 - Experiment with learned encoder instead of PCA to capture nonlinear structure.
 - Compare and contrast different architecture solutions (tabDDPM vs CTGAN)
- 2. Deployment and Production
 - Build FASTAPI endpoint or develop lightweight UI Dashboard (streamlit) where users can synthesize cohorts, visualize distributions and validate outputs.